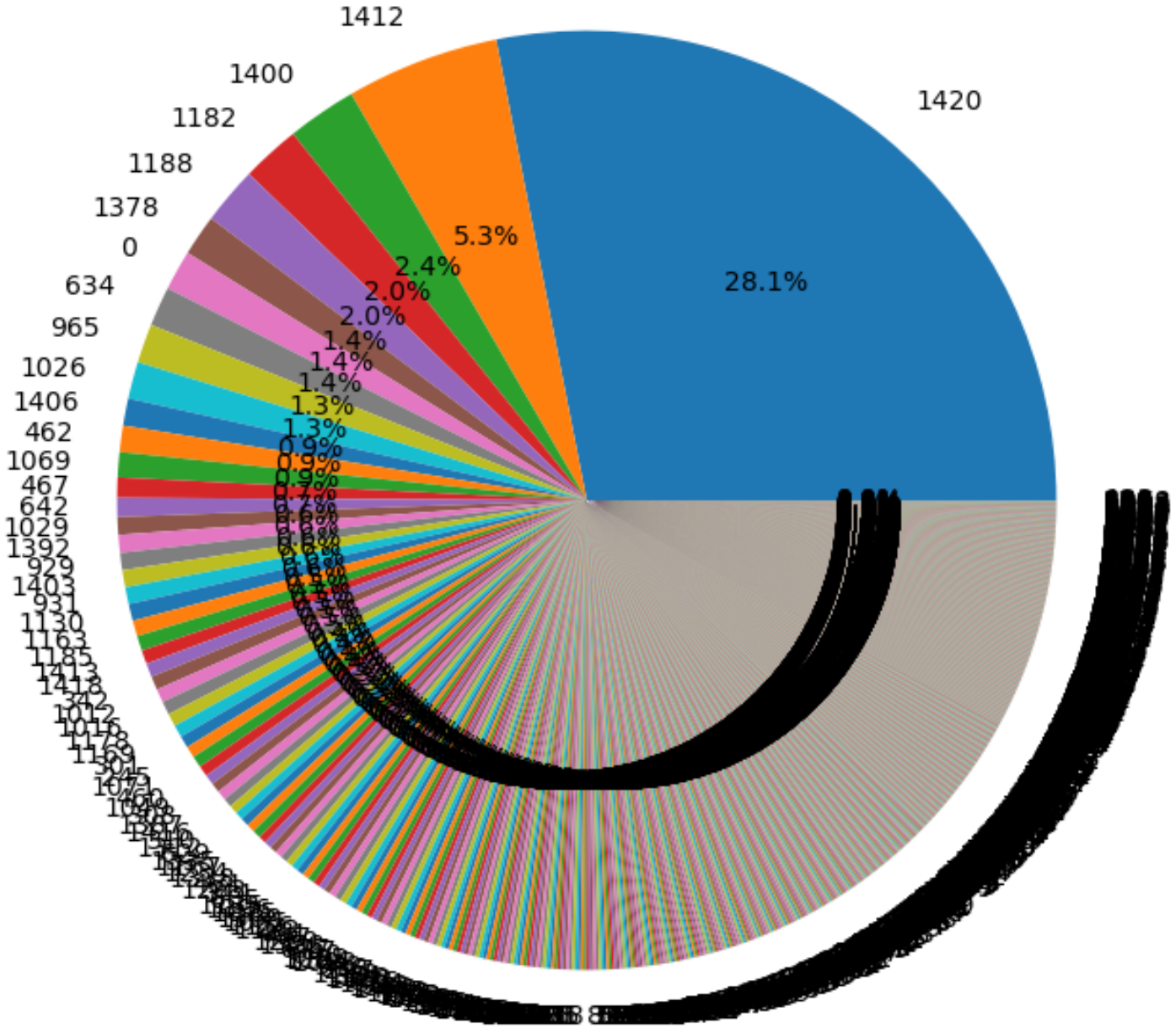
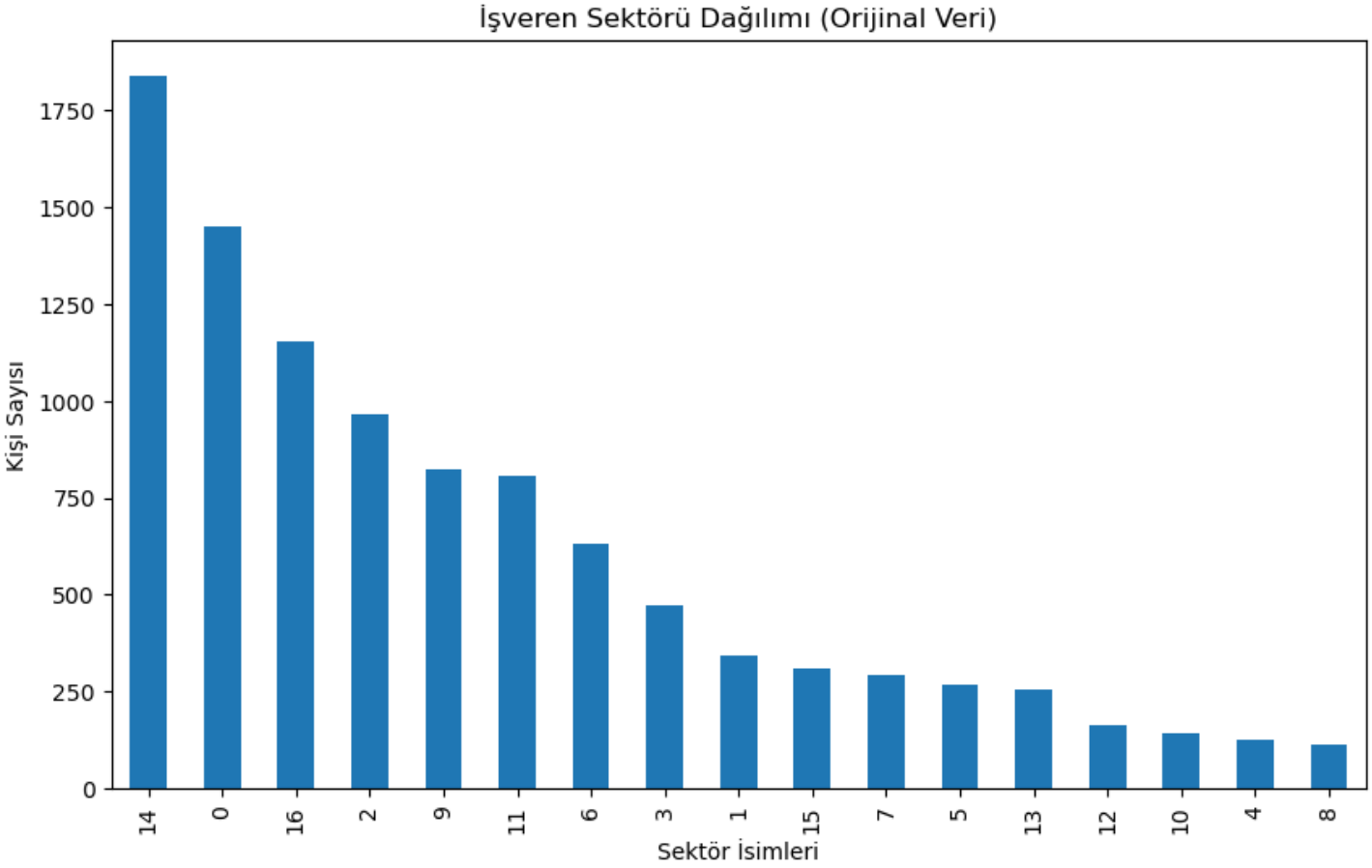
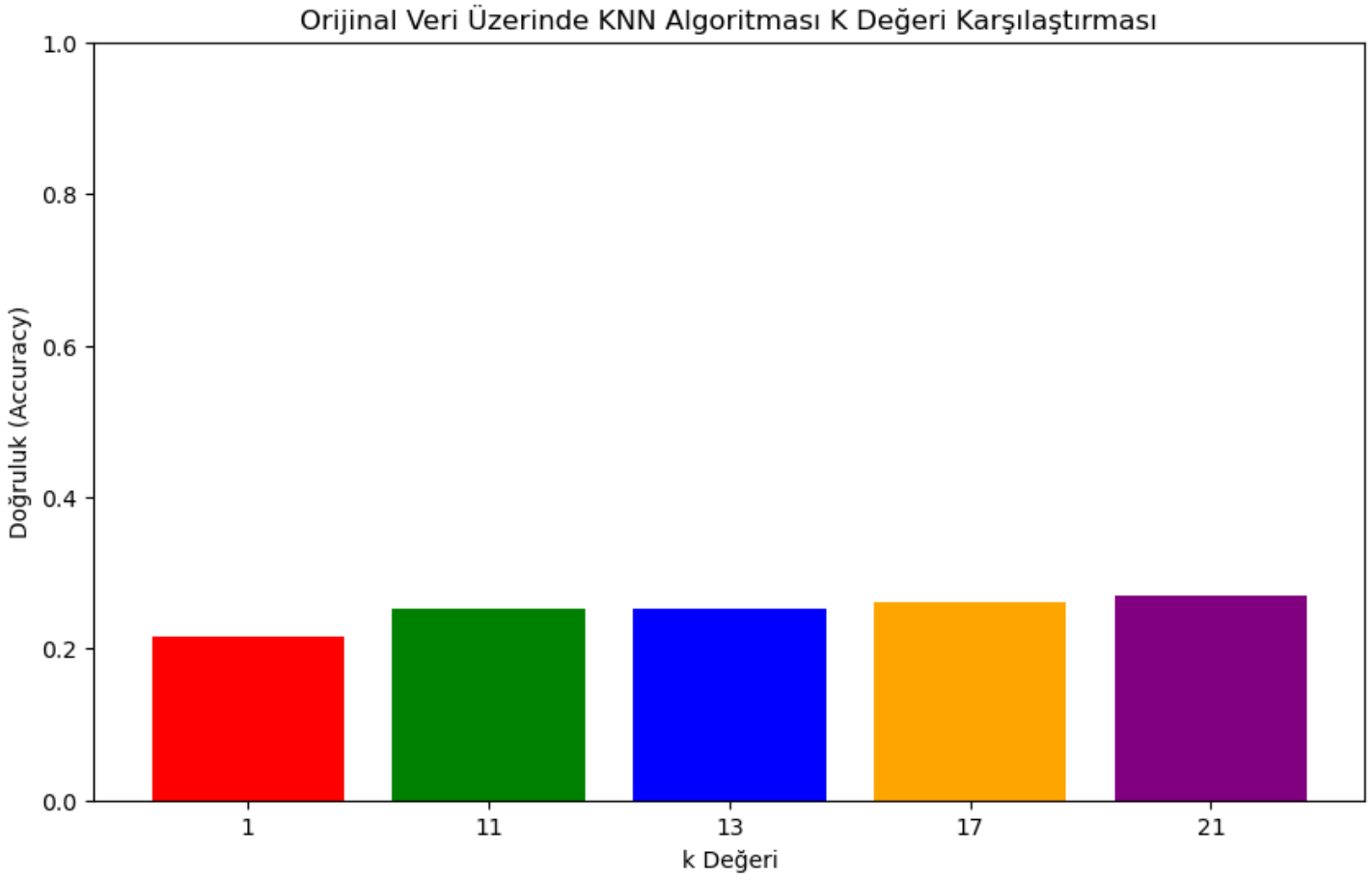
Veri Madenciligi Grafik ve Matrisler (Orijinal ve Onislenmis Veri)

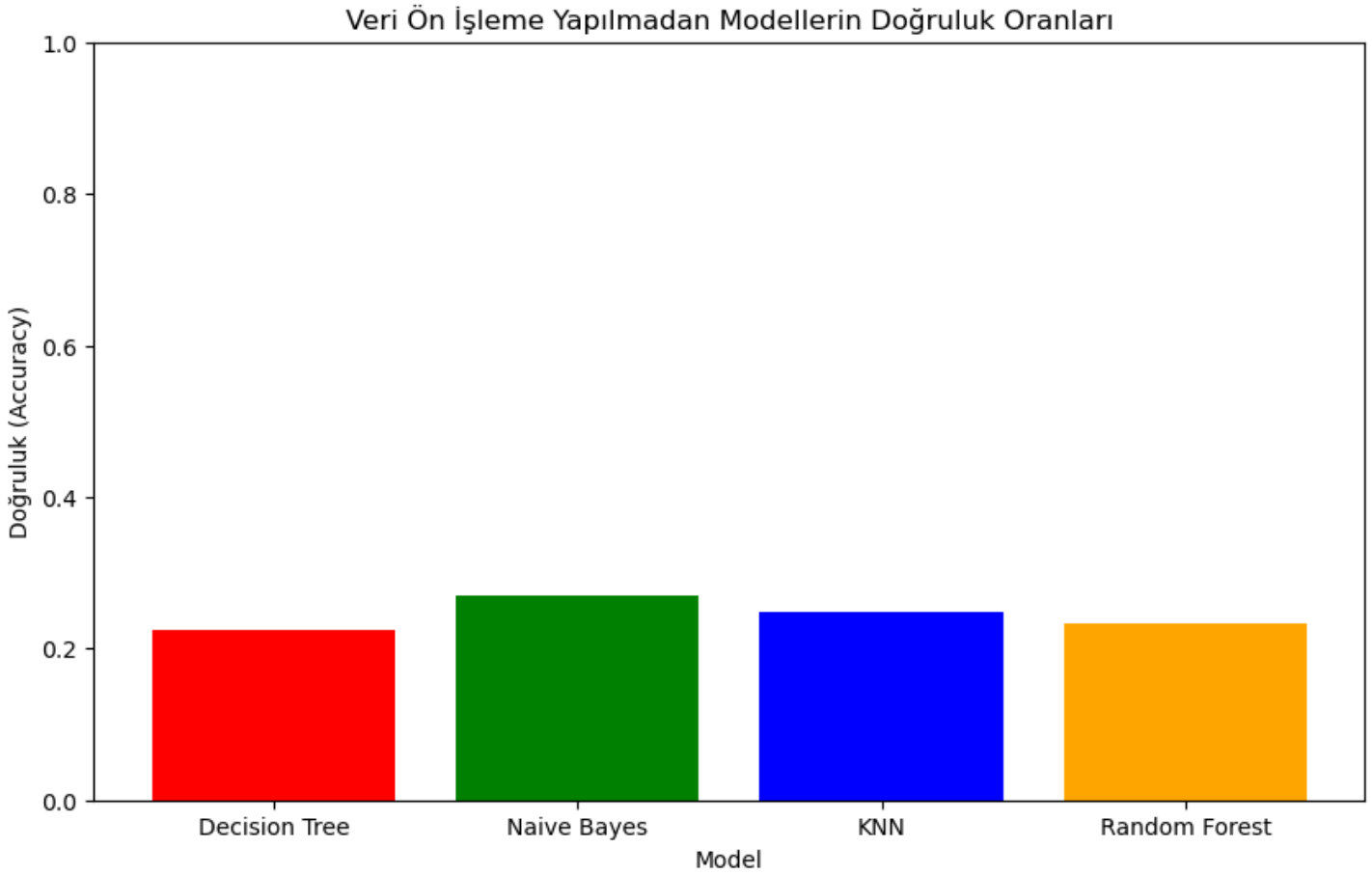
Orijinal Veri Gorselleri

Çalışan Algoritmaların Dağılımı (Orijinal Veri)

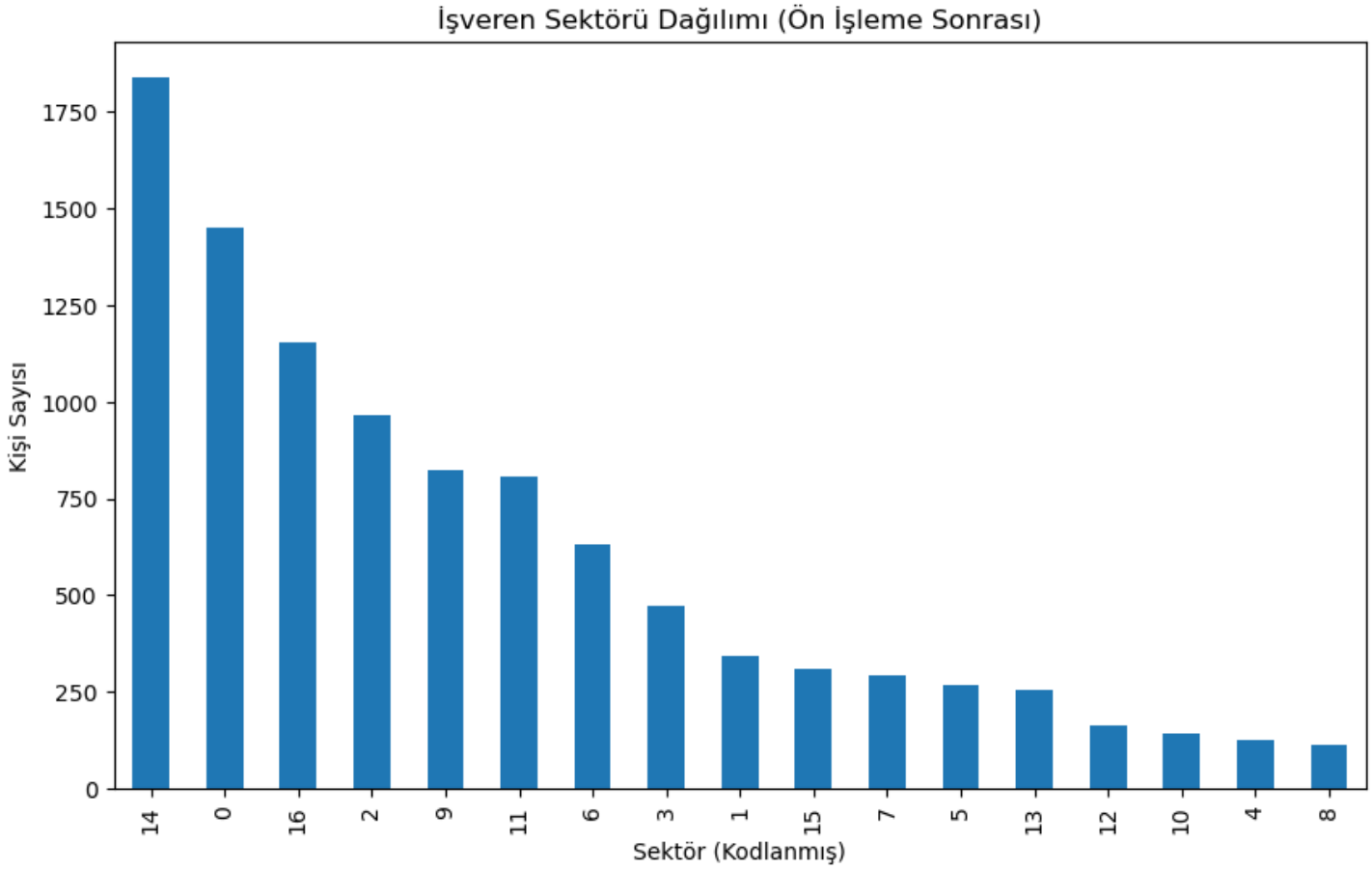



İşveren Sektörü Dağılımı (Orijinal Veri)
Kişi Sayısı
0
250
500
750
1000
1250
1500
1750
14
0
16
2
9
11
6
3
1
15
7
5
13
12
10
4
8
Sektör İsimleri


Orijinal Veri Üzerinde KNN Algoritması K Değeri Karşılaştırması
Doğruluk (Accuracy)
1.0
0.8
0.6
0.4
0.2
0.0
1
11
13
17
21
k Değeri


Veri Ön İşleme Yapılmadan Modellerin Doğruluk Oranları
1.0
0.8
0.6
0.4
0.2
0.0
Doğruluk (Accuracy)
Decision Tree
Naive Bayes
KNN
Random Forest
Model

Onislenmis Veri Gorselleri


İşveren Sektörü Dağılımı (Ön İşleme Sonrası)
Kişi Sayısı
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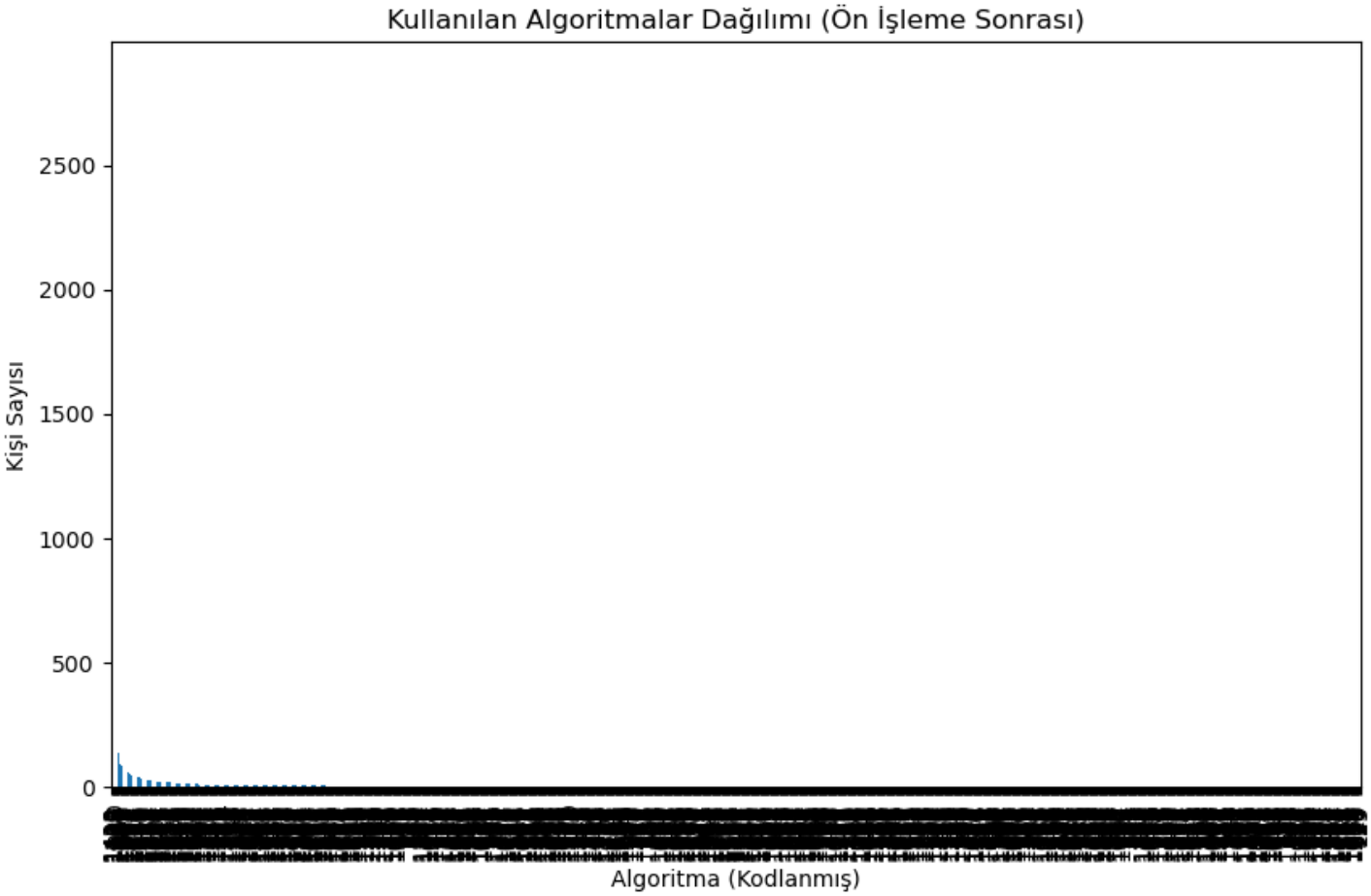
250
500
750
1000
1250
1500
1750
14
0
16
2
9
11
6
3
1
15
7
5
13
12
10
4
8
Sektör (Kodlanmış)


Kullanılan Algoritmalar Dağılımı (Ön İşleme Sonrası)
Kişi Sayısı
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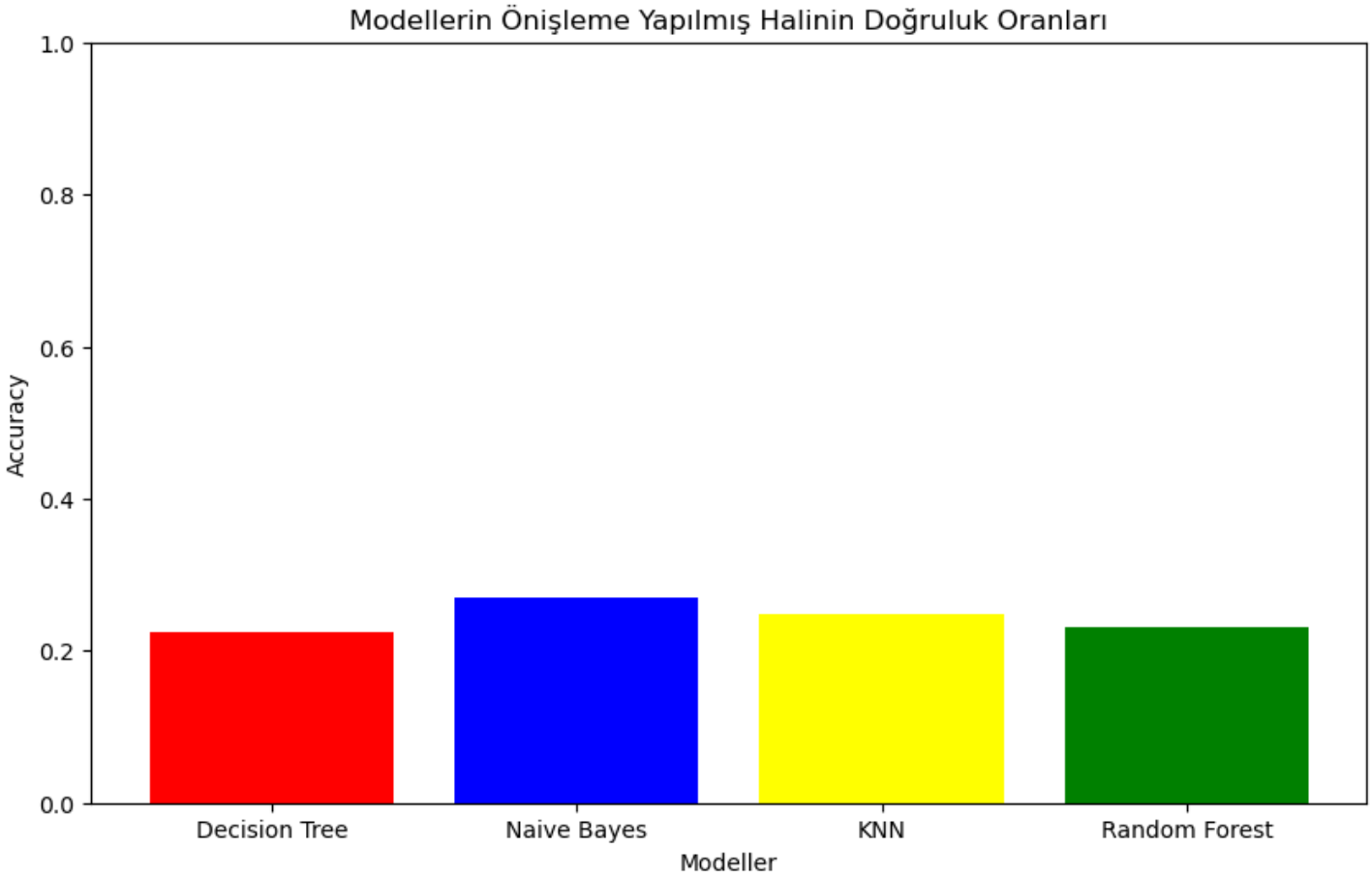
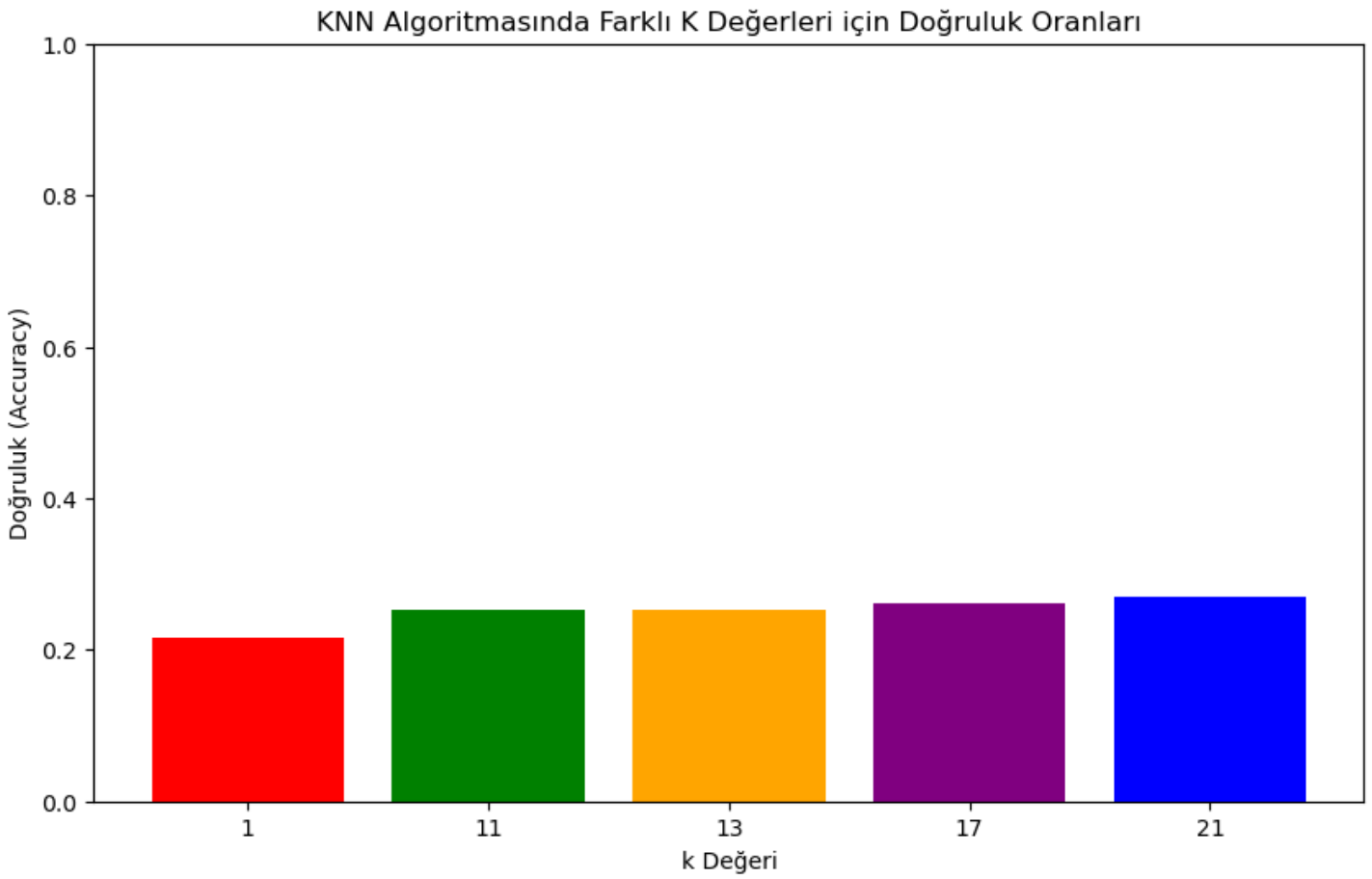
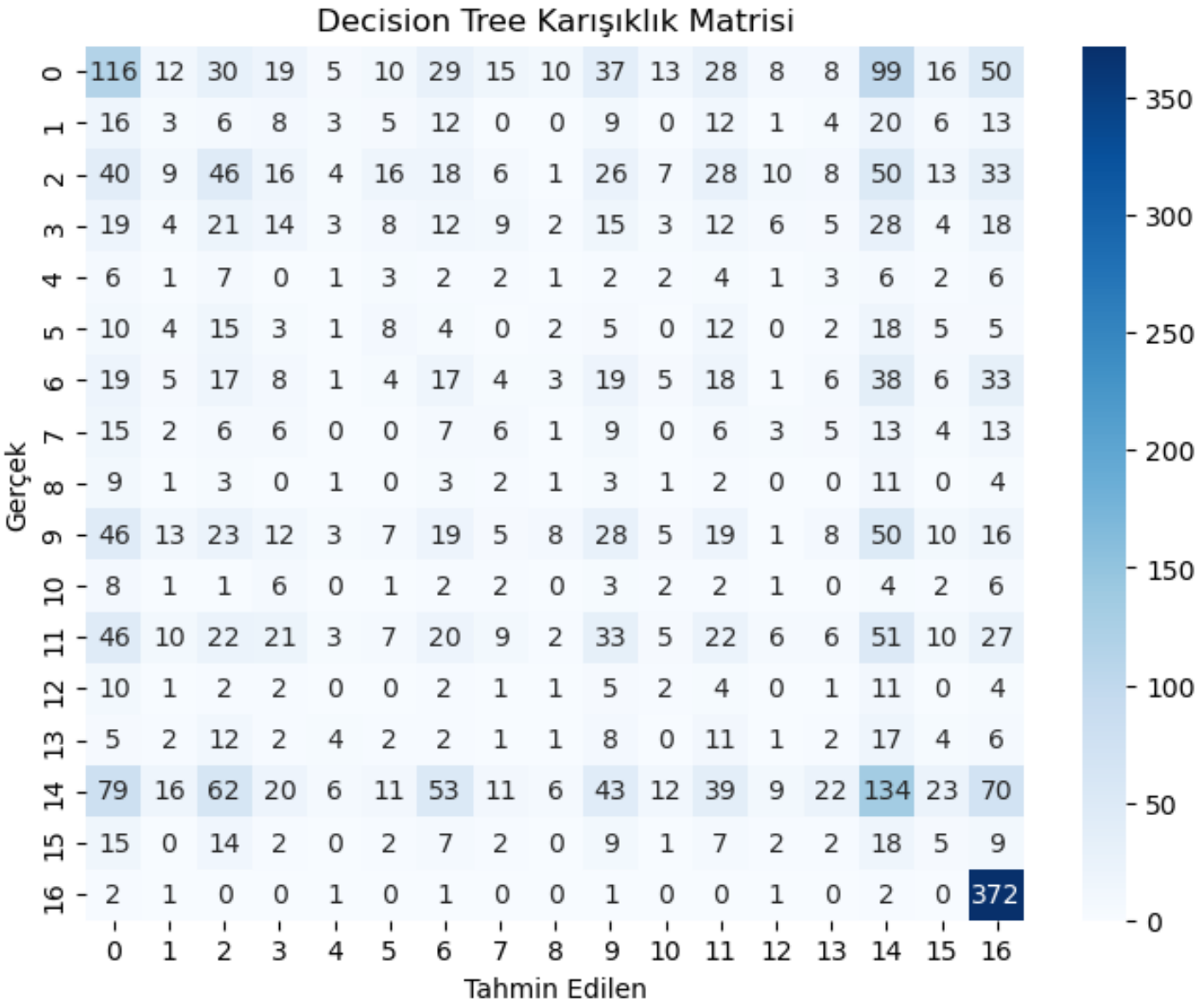
1500
1000
500
0
Algoritma (Kodlanmış)


Modellerin Önişleme Yapılmış Halinin Doğruluk Oranları
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
Decision Tree
Naive Bayes
KNN
Random Forest
Modeller


KNN Algoritmasında Farklı K Değerleri için Doğruluk Oranları
1.0
0.8
0.6
0.4
0.2
0.0
Doğruluk (Accuracy)
1
11
13
17
21
k Değeri


Decision Tree Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 116 12 30 19 5 10 29 15 10 37 13 28 8 8 99 16 50
1 16 3 6 8 3 5 12 0 0 9 0 12 1 4 20 6 13
2 40 9 46 16 4 16 18 6 1 26 7 28 10 8 50 13 33
3 19 4 21 14 3 8 12 9 2 15 3 12 6 5 28 4 18
4 6 1 7 0 1 3 2 2 1 2 2 4 1 3 6 2 6
5 10 4 15 3 1 8 4 0 2 5 0 12 0 2 18 5 5
6 19 5 17 8 1 4 17 4 3 19 5 18 1 6 38 6 33
7 15 2 6 6 0 0 7 6 1 9 0 6 3 5 13 4 13
8 9 1 3 0 1 0 3 2 1 3 1 2 0 0 11 0 4
9 46 13 23 12 3 7 19 5 8 28 5 19 1 8 50 10 16
10 8 1 1 6 0 1 2 2 0 3 2 2 1 0 4 2 6
11 46 10 22 21 3 7 20 9 2 33 5 22 6 6 51 10 27
12 10 1 2 2 0 0 2 1 1 5 2 4 0 1 11 0 4
13 5 2 12 2 4 2 2 1 1 8 0 11 1 2 17 4 6
14 79 16 62 20 6 11 53 11 6 43 12 39 9 22 134 23 70
15 15 0 14 2 0 2 7 2 0 9 1 7 2 2 18 5 9
16 2 1 0 0 1 0 1 0 0 1 0 0 1 0 2 0 372
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
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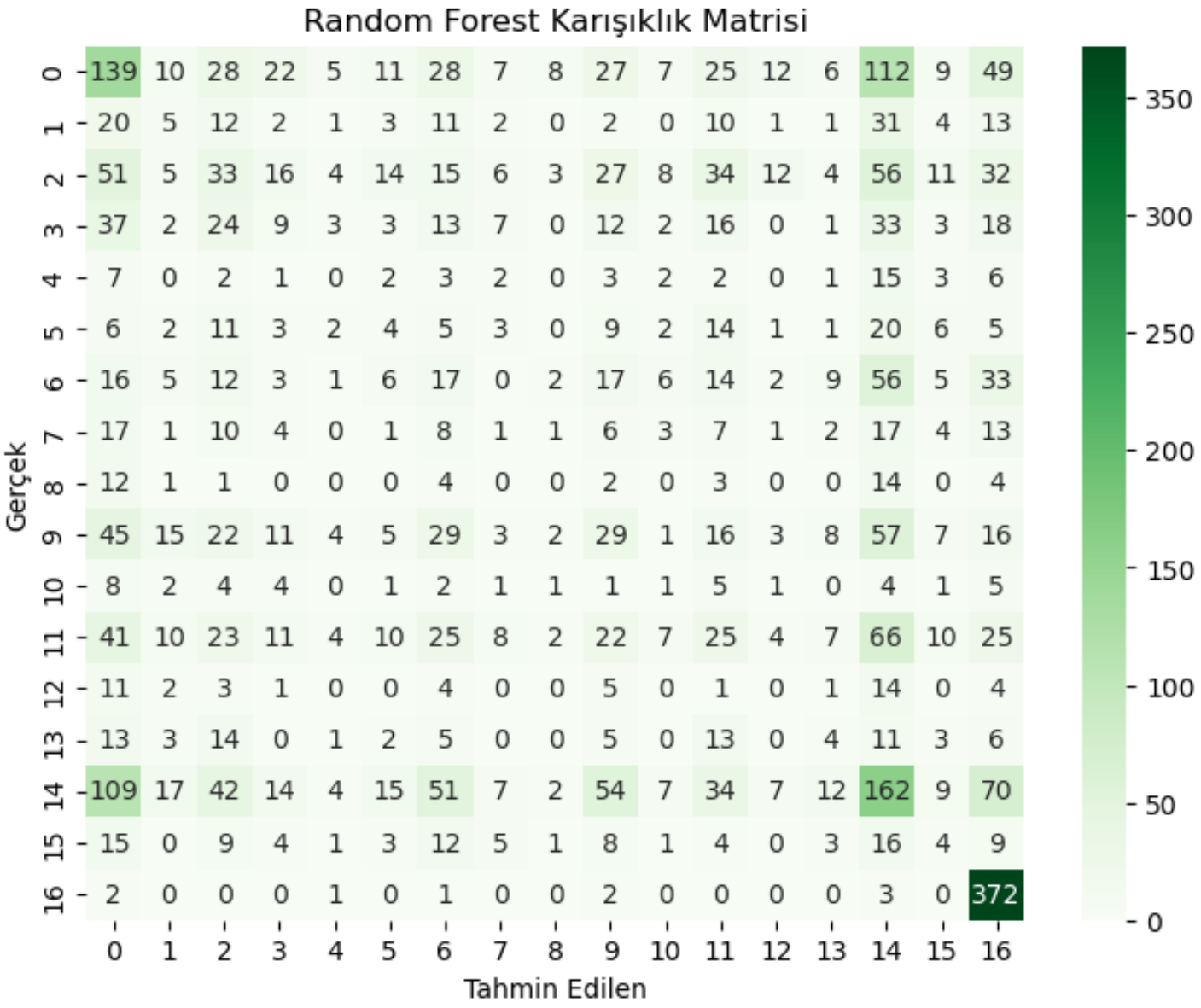
300
250
200
150
100
50
0


Random Forest Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
0 139 10 28 22 5 11 28 7 8 27 7 25 12 6 112 9 49
1 20 5 12 2 1 3 11 2 0 2 0 10 1 1 31 4 13
2 51 5 33 16 4 14 15 6 3 27 8 34 12 4 56 11 32
3 37 2 24 9 3 3 13 7 0 12 2 16 0 1 33 3 18
4 7 0 2 1 0 2 3 2 0 3 2 2 0 1 15 3 6
5 6 2 11 3 2 4 5 3 0 9 2 14 1 1 20 6 5
6 16 5 12 3 1 6 17 0 2 17 6 14 2 9 56 5 33
7 17 1 10 4 0 1 8 1 1 6 3 7 1 2 17 4 13
8 12 1 1 0 0 0 4 0 0 2 0 3 0 0 14 0 4
9 45 15 22 11 4 5 29 3 2 29 1 16 3 8 57 7 16
10 8 2 4 4 0 1 2 1 1 1 1 5 1 0 4 1 5
11 41 10 23 11 4 10 25 8 2 22 7 25 4 7 66 10 25
12 11 2 3 1 0 0 4 0 0 5 0 1 0 1 14 0 4
13 13 3 14 0 1 2 5 0 0 5 0 13 0 4 11 3 6
14 109 17 42 14 4 15 51 7 2 54 7 34 7 12 162 9 70
15 15 0 9 4 1 3 12 5 1 8 1 4 0 3 16 4 9
16 2 0 0 0 1 0 1 0 0 2 0 0 0 0 3 0 372
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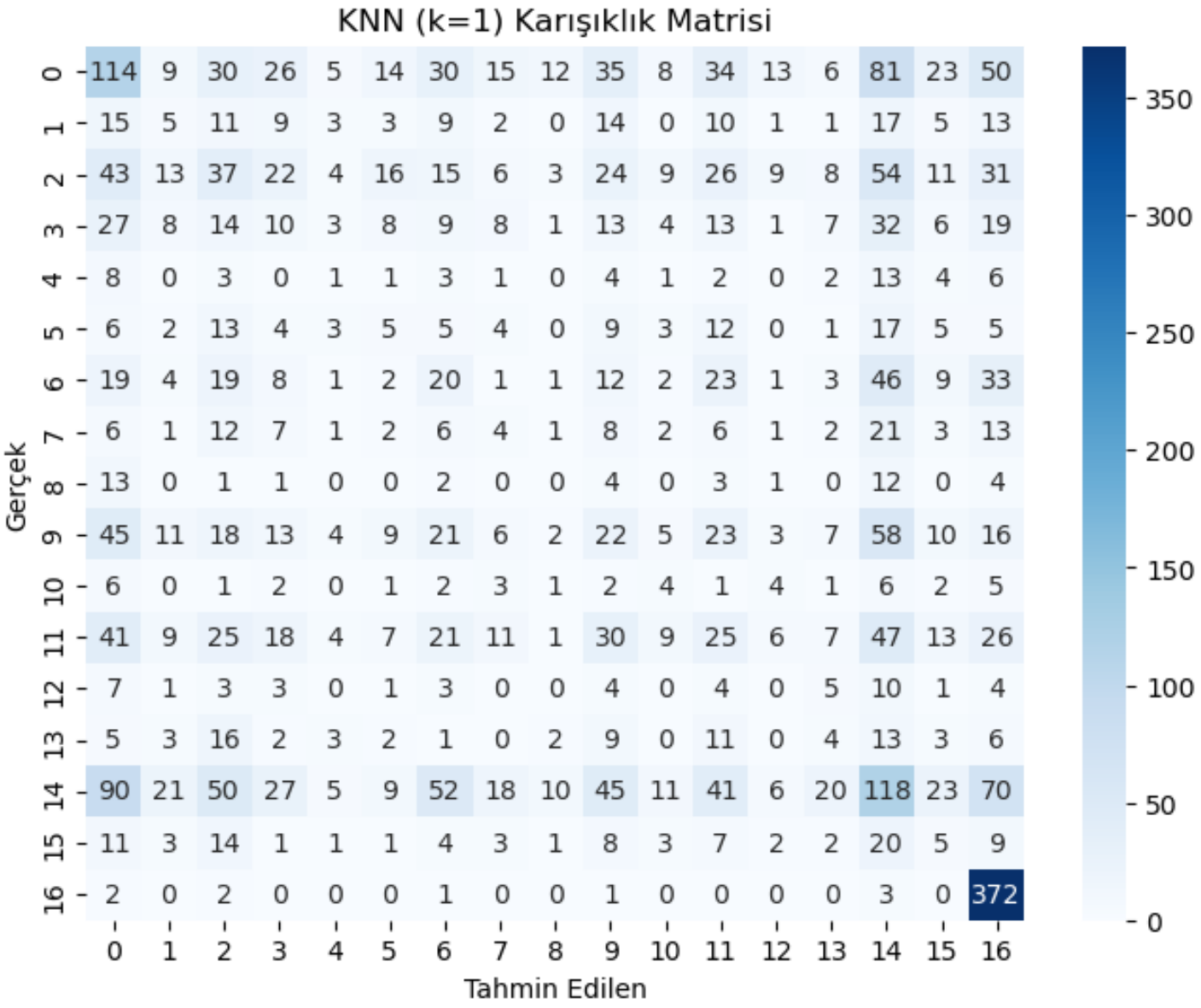
200
150
100
50
0


KNN (k=1) Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 114 9 30 26 5 14 30 15 12 35 8 34 13 6 81 23 50
1 15 5 11 9 3 3 9 2 0 14 0 10 1 1 17 5 13
2 43 13 37 22 4 16 15 6 3 24 9 26 9 8 54 11 31
3 27 8 14 10 3 8 9 8 1 13 4 13 1 7 32 6 19
4 8 0 3 0 1 1 3 1 0 4 1 2 0 2 13 4 6
5 6 2 13 4 3 5 5 4 0 9 3 12 0 1 17 5 5
6 19 4 19 8 1 2 20 1 1 12 2 23 1 3 46 9 33
7 6 1 12 7 1 2 6 4 1 8 2 6 1 2 21 3 13
8 13 0 1 1 0 0 2 0 0 4 0 3 1 0 12 0 4
9 45 11 18 13 4 9 21 6 2 22 5 23 3 7 58 10 16
10 6 0 1 2 0 1 2 3 1 2 4 1 4 1 6 2 5
11 41 9 25 18 4 7 21 11 1 30 9 25 6 7 47 13 26
12 7 1 3 3 0 1 3 0 0 4 0 4 0 5 10 1 4
13 5 3 16 2 3 2 1 0 2 9 0 11 0 4 13 3 6
14 90 21 50 27 5 9 52 18 10 45 11 41 6 20 118 23 70
15 11 3 14 1 1 1 4 3 1 8 3 7 2 2 20 5 9
16 2 0 2 0 0 0 1 0 0 1 0 0 0 0 3 0 372
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
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300
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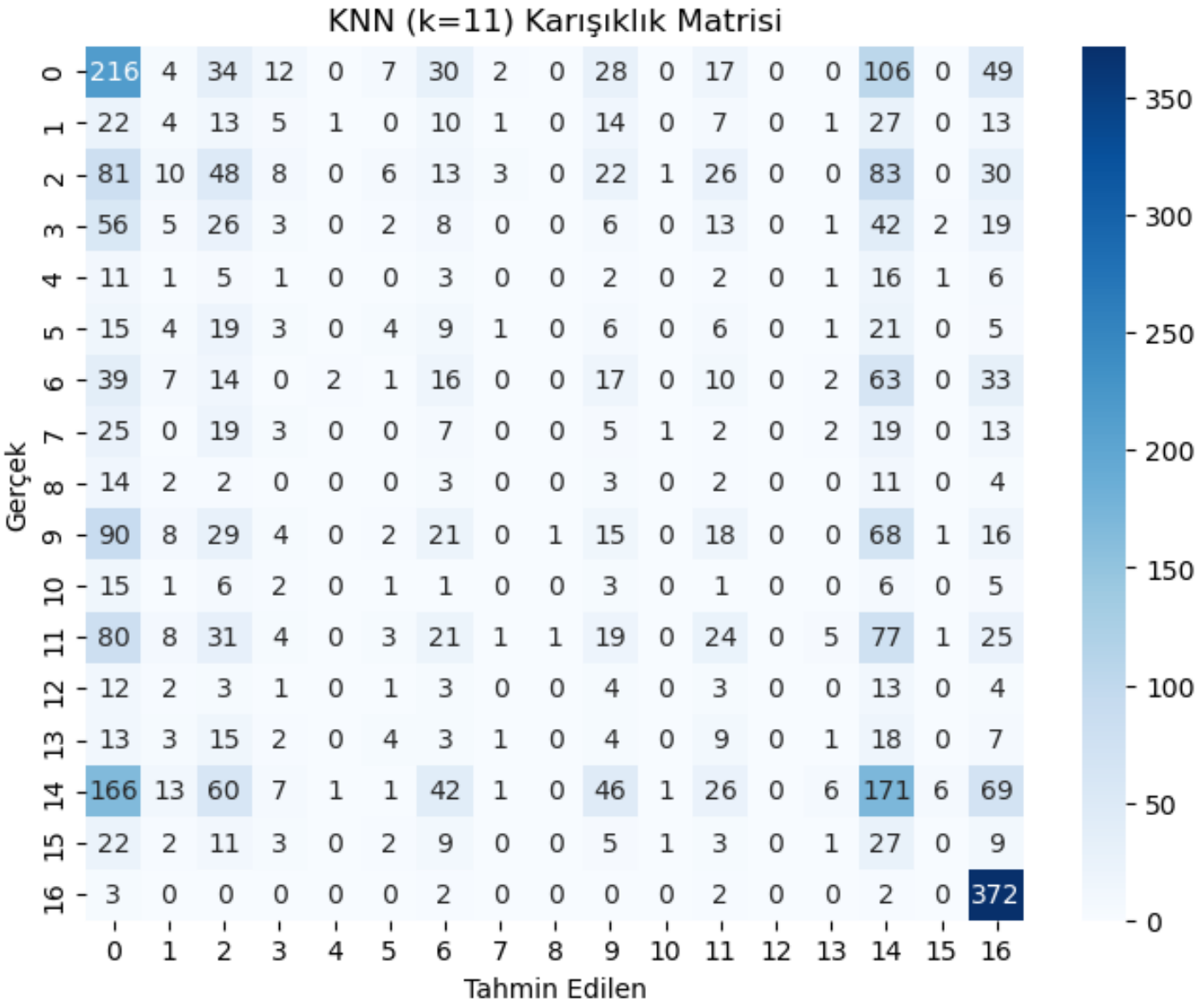
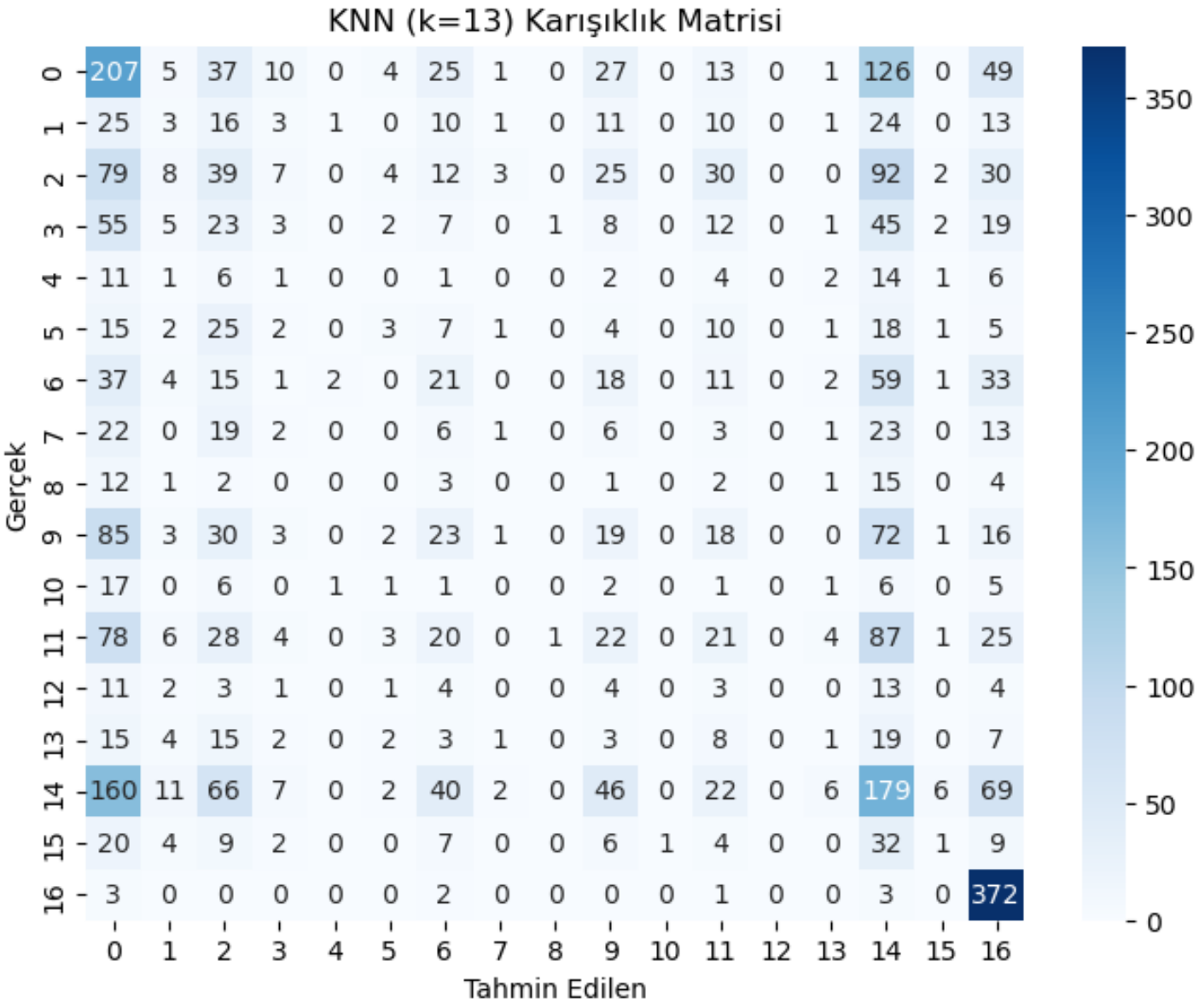
200
150
100
50
0


KNN (k=11) Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
0 216 4 34 12 0 7 30 2 0 28 0 17 0 0 106 0 49
1 22 4 13 5 1 0 10 1 0 14 0 7 0 1 27 0 13
2 81 10 48 8 0 6 13 3 0 22 1 26 0 0 83 0 30
3 56 5 26 3 0 2 8 0 0 6 0 13 0 1 42 2 19
4 11 1 5 1 0 0 3 0 0 2 0 2 0 1 16 1 6
5 15 4 19 3 0 4 9 1 0 6 0 6 0 1 21 0 5
6 39 7 14 0 2 1 16 0 0 17 0 10 0 2 63 0 33
7 25 0 19 3 0 0 7 0 0 5 1 2 0 2 19 0 13
8 14 2 2 0 0 0 3 0 0 3 0 2 0 0 11 0 4
9 90 8 29 4 0 2 21 0 1 15 0 18 0 0 68 1 16
10 15 1 6 2 0 1 1 0 0 3 0 1 0 0 6 0 5
11 80 8 31 4 0 3 21 1 1 19 0 24 0 5 77 1 25
12 12 2 3 1 0 1 3 0 0 4 0 3 0 0 13 0 4
13 13 3 15 2 0 4 3 1 0 4 0 9 0 1 18 0 7
14 166 13 60 7 1 1 42 1 0 46 1 26 0 6 171 6 69
15 22 2 11 3 0 2 9 0 0 5 1 3 0 1 27 0 9
16 3 0 0 0 0 0 2 0 0 0 0 2 0 0 2 0 372
350
300
250
200
150
100
50
0


KNN (k=13) Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
0 207 5 37 10 0 4 25 1 0 27 0 13 0 1 126 0 49
1 25 3 16 3 1 0 10 1 0 11 0 10 0 1 24 0 13
2 79 8 39 7 0 4 12 3 0 25 0 30 0 0 92 2 30
3 55 5 23 3 0 2 7 0 1 8 0 12 0 1 45 2 19
4 11 1 6 1 0 0 1 0 0 2 0 4 0 2 14 1 6
5 15 2 25 2 0 3 7 1 0 4 0 10 0 1 18 1 5
6 37 4 15 1 2 0 21 0 0 18 0 11 0 2 59 1 33
7 22 0 19 2 0 0 6 1 0 6 0 3 0 1 23 0 13
8 12 1 2 0 0 0 3 0 0 1 0 2 0 1 15 0 4
9 85 3 30 3 0 2 23 1 0 19 0 18 0 0 72 1 16
10 17 0 6 0 1 1 1 0 0 2 0 1 0 1 6 0 5
11 78 6 28 4 0 3 20 0 1 22 0 21 0 4 87 1 25
12 11 2 3 1 0 1 4 0 0 4 0 3 0 0 13 0 4
13 15 4 15 2 0 2 3 1 0 3 0 8 0 1 19 0 7
14 160 11 66 7 0 2 40 2 0 46 0 22 0 6 179 6 69
15 20 4 9 2 0 0 7 0 0 6 1 4 0 0 32 1 9
16 3 0 0 0 0 0 2 0 0 0 0 1 0 0 3 0 372
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300
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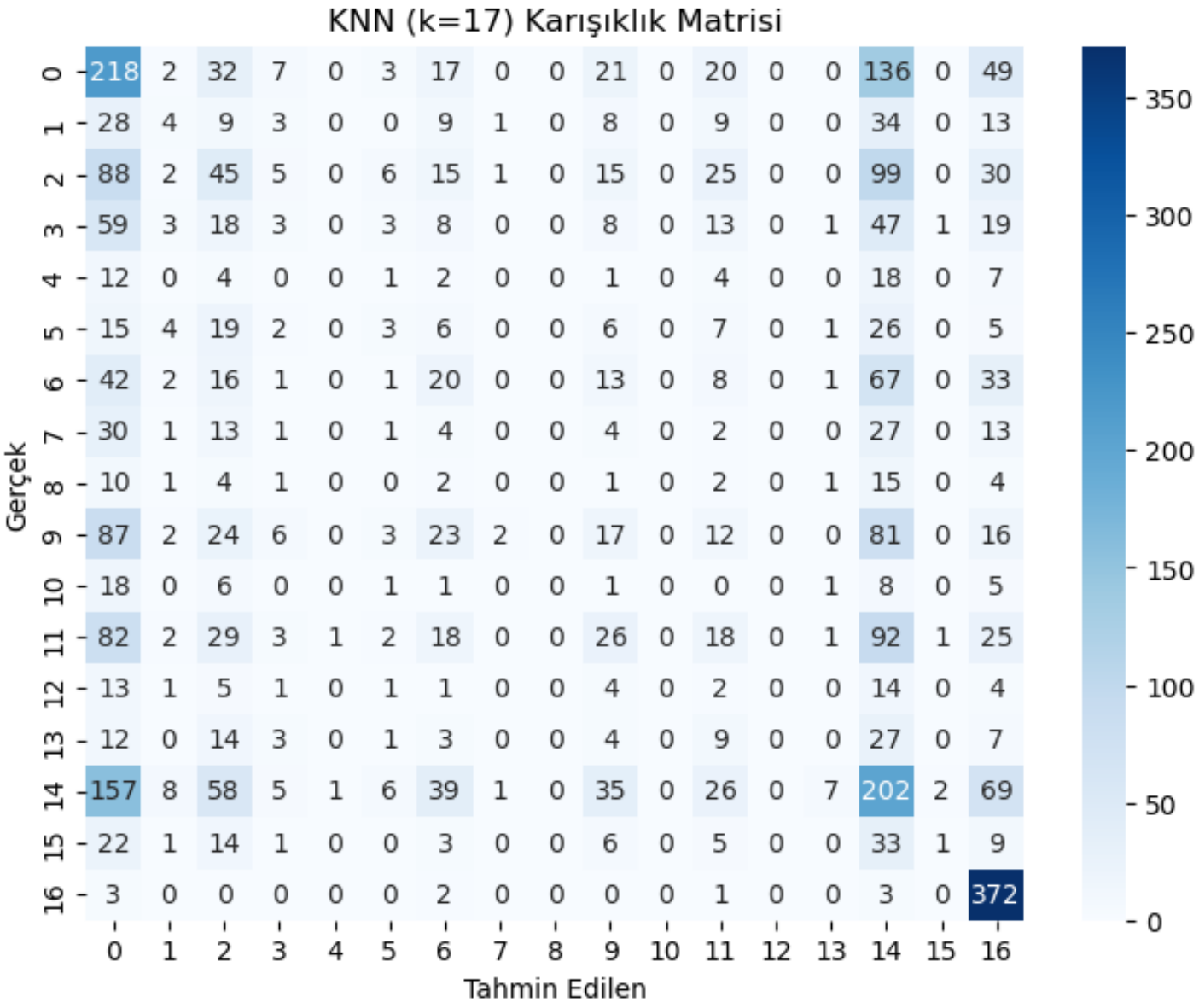
200
150
100
50
0


KNN (k=17) Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
0 218 2 32 7 0 3 17 0 0 21 0 20 0 0 136 0 49
1 28 4 9 3 0 0 9 1 0 8 0 9 0 0 34 0 13
2 88 2 45 5 0 6 15 1 0 15 0 25 0 0 99 0 30
3 59 3 18 3 0 3 8 0 0 8 0 13 0 1 47 1 19
4 12 0 4 0 0 1 2 0 0 1 0 4 0 0 18 0 7
5 15 4 19 2 0 3 6 0 0 6 0 7 0 1 26 0 5
6 42 2 16 1 0 1 20 0 0 13 0 8 0 1 67 0 33
7 30 1 13 1 0 1 4 0 0 4 0 2 0 0 27 0 13
8 10 1 4 1 0 0 2 0 0 1 0 2 0 1 15 0 4
9 87 2 24 6 0 3 23 2 0 17 0 12 0 0 81 0 16
10 18 0 6 0 0 1 1 0 0 1 0 0 0 1 8 0 5
11 82 2 29 3 1 2 18 0 0 26 0 18 0 1 92 1 25
12 13 1 5 1 0 1 1 0 0 4 0 2 0 0 14 0 4
13 12 0 14 3 0 1 3 0 0 4 0 9 0 0 27 0 7
14 157 8 58 5 1 6 39 1 0 35 0 26 0 7 202 2 69
15 22 1 14 1 0 0 3 0 0 6 0 5 0 0 33 1 9
16 3 0 0 0 0 0 2 0 0 0 0 1 0 0 3 0 372
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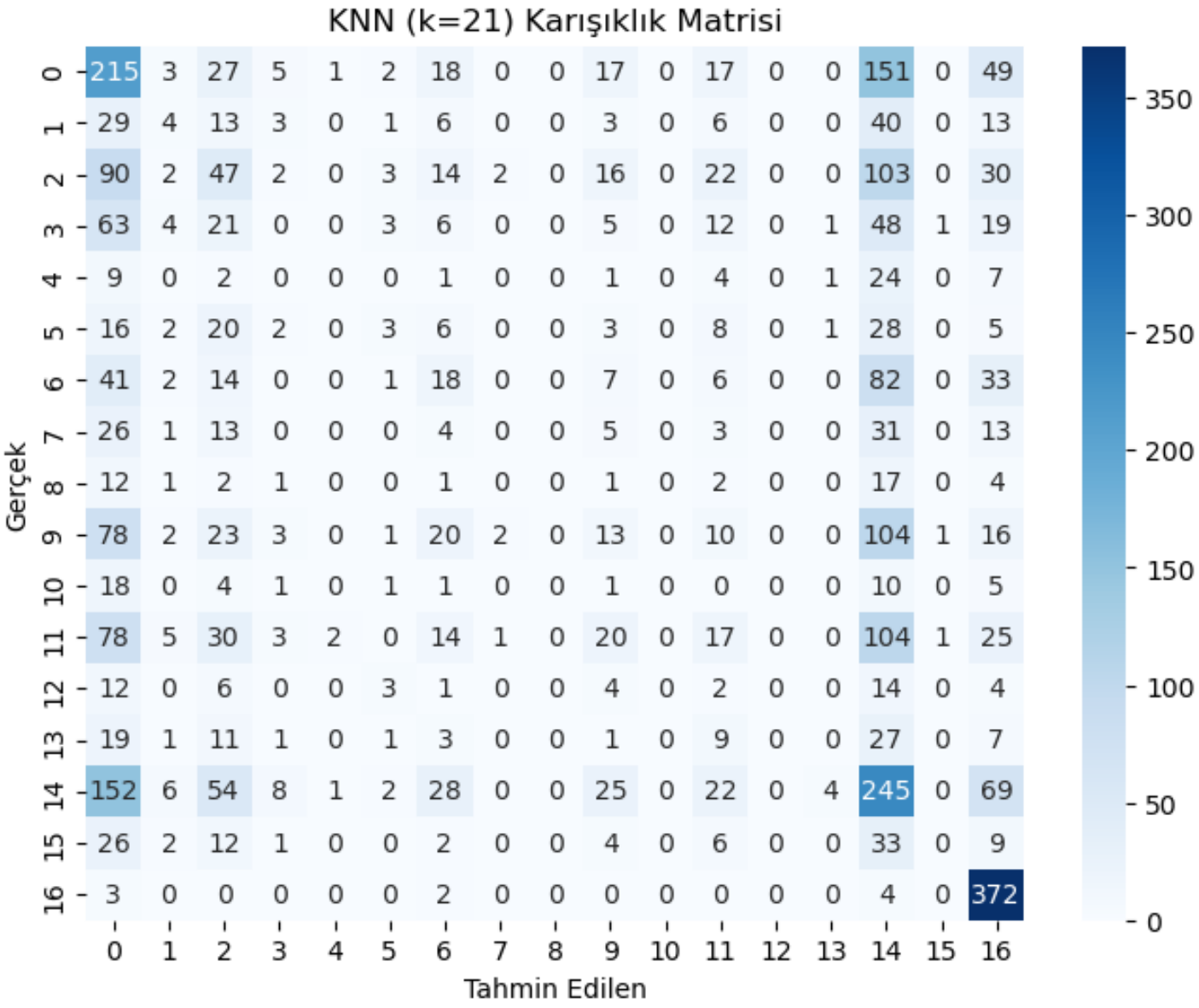
50
0


KNN (k=21) Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 215 3 27 5 1 2 18 0 0 17 0 17 0 0 151 0 49
1 29 4 13 3 0 1 6 0 0 3 0 6 0 0 40 0 13
2 90 2 47 2 0 3 14 2 0 16 0 22 0 0 103 0 30
3 63 4 21 0 0 3 6 0 0 5 0 12 0 1 48 1 19
4 9 0 2 0 0 0 1 0 0 1 0 4 0 1 24 0 7
5 16 2 20 2 0 3 6 0 0 3 0 8 0 1 28 0 5
6 41 2 14 0 0 1 18 0 0 7 0 6 0 0 82 0 33
7 26 1 13 0 0 0 4 0 0 5 0 3 0 0 31 0 13
8 12 1 2 1 0 0 1 0 0 1 0 2 0 0 17 0 4
9 78 2 23 3 0 1 20 2 0 13 0 10 0 0 104 1 16
10 18 0 4 1 0 1 1 0 0 1 0 0 0 0 10 0 5
11 78 5 30 3 2 0 14 1 0 20 0 17 0 0 104 1 25
12 12 0 6 0 0 3 1 0 0 4 0 2 0 0 14 0 4
13 19 1 11 1 0 1 3 0 0 1 0 9 0 0 27 0 7
14 152 6 54 8 1 2 28 0 0 25 0 22 0 4 245 0 69
15 26 2 12 1 0 0 2 0 0 4 0 6 0 0 33 0 9
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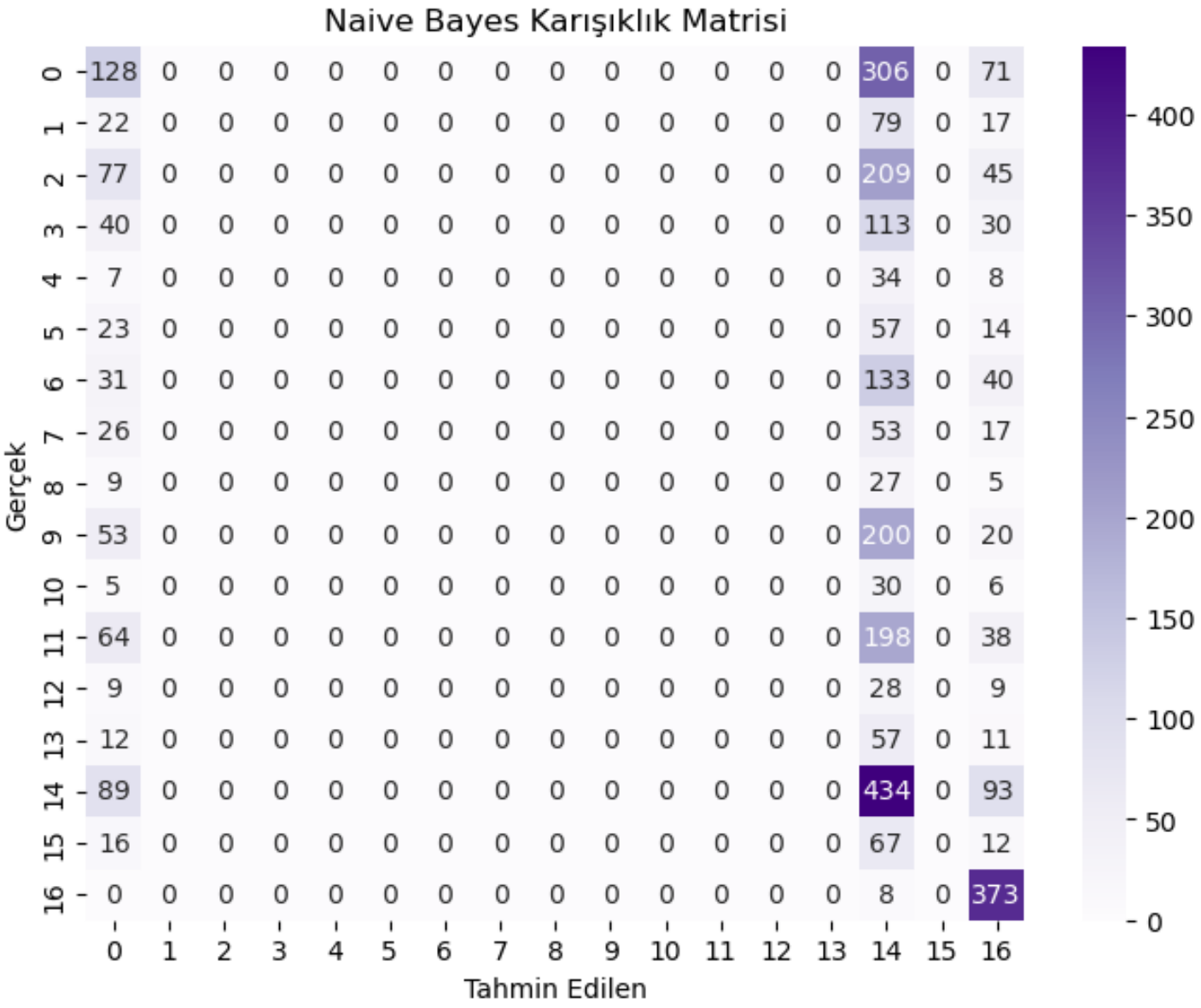
50
0


Naive Bayes Karışıklık Matrisi
Gerçek
Tahmin Edilen
0 128 0 0 0 0 0 0 0 0 0 0 0 0 0 306 0 71
1 22 0 0 0 0 0 0 0 0 0 0 0 0 0 79 0 17
2 77 0 0 0 0 0 0 0 0 0 0 0 0 0 209 0 45
3 40 0 0 0 0 0 0 0 0 0 0 0 0 0 113 0 30
4 7 0 0 0 0 0 0 0 0 0 0 0 0 0 34 0 8
5 23 0 0 0 0 0 0 0 0 0 0 0 0 0 57 0 14
6 31 0 0 0 0 0 0 0 0 0 0 0 0 0 133 0 40
7 26 0 0 0 0 0 0 0 0 0 0 0 0 0 53 0 17
8 9 0 0 0 0 0 0 0 0 0 0 0 0 0 27 0 5
9 53 0 0 0 0 0 0 0 0 0 0 0 0 0 200 0 20
10 5 0 0 0 0 0 0 0 0 0 0 0 0 0 30 0 6
11 64 0 0 0 0 0 0 0 0 0 0 0 0 0 198 0 38
12 9 0 0 0 0 0 0 0 0 0 0 0 0 0 28 0 9
13 12 0 0 0 0 0 0 0 0 0 0 0 0 0 57 0 11
14 89 0 0 0 0 0 0 0 0 0 0 0 0 0 434 0 93
15 16 0 0 0 0 0 0 0 0 0 0 0 0 0 67 0 12
16 0 0 0 0 0 0 0 0 0 0 0 0 0 0 8 0 373
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
400
350
300
250
200
150
100
50
0